Day 25 – Basic Incident Response Report

Incident response is a structured process used to identify, manage, contain, and recover from cybersecurity incidents. A security incident may involve malware infections, unauthorized access, phishing attacks, data breaches, or suspicious network activity.

An effective incident response process helps organizations minimize damage, protect sensitive information, reduce operational downtime, and restore affected systems safely. It also helps security teams understand how an incident occurred and implement improvements to prevent similar incidents in the future.

Objective

The objective of this task is to understand the fundamental stages of incident response and develop a practical response checklist. The task focuses on preparation, detection and analysis, containment, eradication, recovery, and lessons learned.

It also aims to explain how organizations coordinate their response activities, preserve evidence, communicate with relevant stakeholders, and strengthen their security practices after an incident.

Tools Used

Google Docs: Used to prepare the incident response checklist and document the incident handling process.

Security Logs: Used to identify suspicious activities, investigate events, and establish the sequence of actions associated with an incident.

Antivirus and Endpoint Detection Tools: Used to identify malicious files, suspicious processes, and potentially compromised devices.

Network Monitoring Tools: Used to investigate unusual network traffic and identify potentially affected systems.

Backup and Recovery Tools: Used to restore affected systems and data after the threat has been contained and removed.

Understanding Incident Response

Incident response refers to the organized approach an organization follows when a cybersecurity incident occurs. It defines the actions security teams take to detect threats, reduce their impact, eliminate the cause, restore normal operations, and learn from the incident.

A documented incident response plan assigns responsibilities, establishes communication procedures, and provides guidance for handling different types of security incidents.

For example, if an employee opens a malicious attachment and the computer begins communicating with an unfamiliar server, the security team must investigate the activity, isolate the affected device if

necessary, preserve relevant evidence, remove the threat, and verify that the device is safe before returning it to service.

Stages of Incident Response

Preparation

Preparation involves establishing the policies, tools, people, and procedures required to handle security incidents effectively. Organizations should define incident response roles, maintain updated security software, create reliable backups, and prepare communication and escalation procedures.

Regular employee awareness training and incident response exercises help ensure that the organization is ready to respond when a threat occurs.

Detection and Analysis

Detection and analysis involve identifying potential security incidents and determining their nature, scope, and severity. Security teams examine system logs, authentication records, endpoint alerts, and network traffic to identify suspicious patterns.

Once an alert is detected, analysts validate the available evidence, identify potentially affected systems, and assess the risk. Accurate timestamps and preserved logs help establish the sequence of events.

Containment

Containment aims to prevent an incident from spreading and reduce its impact. Depending on the situation, security teams may isolate an affected device from the network, disable a compromised account, block malicious network traffic, or restrict access to affected systems.

Containment decisions should consider operational requirements and the need to preserve evidence. Actions should follow the organization's incident response procedures.

Eradication

Eradication involves removing the underlying cause of the incident. This may include deleting malicious files, removing unauthorized accounts, disabling compromised credentials, correcting vulnerabilities, and eliminating unauthorized access mechanisms.

Before proceeding, security teams should investigate the incident sufficiently to avoid leaving the original cause unresolved.

Recovery

Recovery involves safely restoring affected systems, services, and data to normal operation. Systems may be restored from trusted backups, patched, reconfigured, and monitored for signs of renewed malicious activity.

Security teams should verify that the threat has been removed and that restored systems are functioning correctly before returning them to regular use.

Lessons Learned

After the incident has been resolved, the organization reviews the response to identify what happened, how effectively the incident was handled, and what improvements are needed.

The review may identify gaps in monitoring, access controls, employee awareness, backup procedures, or incident response coordination. Findings should be documented and used to improve security policies, response procedures, and future training.

Incident Response Checklist

The following checklist provides a practical guide for handling a cybersecurity incident.

Preparation

- Establish an incident response team and assign responsibilities.
- Maintain an incident response plan with clear escalation procedures.
- Ensure that security monitoring tools and logging are configured appropriately.
- Maintain tested backups of important systems and data.
- Train employees to recognize and report suspicious activities.

Detection and Analysis

- Identify the alert or event that may indicate an incident.
- Record the detection time and relevant details.
- Review authentication logs, system logs, and security alerts.
- Identify potentially affected users, devices, accounts, and services.
- Assess the severity and potential impact of the incident.
- Preserve relevant evidence and maintain appropriate records.

Containment

- Isolate affected devices or systems when necessary.
- Disable or secure compromised accounts and credentials.
- Block confirmed malicious indicators where appropriate.
- Prevent further unauthorized access or spread.
- Document containment actions and their impact.

Eradication

- Identify and remove malicious files, processes, or unauthorized access mechanisms.
- Address the vulnerabilities or weaknesses that enabled the incident.
- Reset compromised credentials and revoke affected sessions where necessary.

- Verify that the original cause has been addressed.

Recovery

- Restore systems and data from trusted sources when required.
- Apply security patches and verify system configurations.
- Test affected services before returning them to normal operation.
- Monitor restored systems for signs of continued or renewed malicious activity.
- Confirm that recovery is complete and document the outcome.

Lessons Learned

- Document the incident timeline, impact, and response actions.
- Identify what worked well and what needs improvement.
- Update security controls and incident response procedures.
- Provide additional training where necessary.
- Record recommendations and assign responsibility for follow-up actions.

Example of Incident Response

Consider a situation in which a security monitoring system detects repeated failed login attempts against an employee account, followed by an unexpected successful login.

During detection and analysis, the security team reviews authentication logs, identifies the source IP address, and checks whether the successful login matches the employee's normal activity.

If the evidence suggests that the account may be compromised, the team proceeds with containment by securing the account, revoking suspicious sessions, and resetting credentials through approved procedures. Relevant logs and other evidence are preserved for investigation.

During eradication, the team investigates how access was obtained and addresses any identified weaknesses. During recovery, the account is restored to normal use with appropriate security protections, such as multifactor authentication, and the activity is monitored.

Finally, the team documents the incident, evaluates the effectiveness of its response, and identifies improvements to reduce the likelihood of similar incidents.

This example illustrates how the stages of incident response work together to manage a potential security threat.

Security Recommendations

Maintain an Incident Response Plan: Keep documented procedures available and review them regularly.

Enable Security Monitoring: Collect and review relevant logs to identify suspicious activities as early as possible.

Use Multifactor Authentication: Add an extra layer of protection to accounts, especially privileged accounts.

Maintain Secure Backups: Store backups securely and test restoration procedures regularly.

Preserve Evidence: Protect relevant logs and records so that incident investigations can establish what happened.

Restrict Access: Apply the principle of least privilege and review account permissions regularly.

Communicate Effectively: Establish clear procedures for notifying the incident response team, management, and other appropriate stakeholders.

Review Every Incident: Use lessons learned to improve monitoring, training, response procedures, and security controls.

Incident response is an essential component of cybersecurity because it provides a structured method for managing security incidents and minimizing their impact. The process includes preparation, detection and analysis, containment, eradication, recovery, and lessons learned.

This task demonstrates the importance of having a clear incident response checklist, preserving evidence, communicating effectively, and verifying that affected systems are safe before restoring normal operations. By preparing in advance and continuously improving response procedures, organizations can strengthen their security posture and respond more effectively to future threats.